

A frontier model from a stack of efficiency levers, not brute scale

DeepSeek-V3 -- 671B total params, 37B active per token

**Inherited from DeepSeek-V2
(thoroughly re-validated in V3)**

MLA -- compress the KV cache

DeepSeekMoE -- fine-grained experts

New in DeepSeek-V3

Auxiliary-loss-free load balancing

FP8 training at large scale

Multi-Token Prediction (MTP)

DualPipe co-design: computation-communication overlap -> a big reason it was cheap

 inherited from V2

 new in V3

schematic